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The Editors
Nature

Dear Editors,

We submit for your consideration our manuscript, “Individual infectiousness timing as a hidden driver of epidemic variability and superspreading.”

A central insight of modern infectious disease epidemiology is that individual-level variation matters: Lloyd-Smith et al. (2005, *Nature*) showed that variation in the *amount* of individual transmission, captured by the dispersion parameter k , profoundly shapes epidemic dynamics, outbreak probability, and the effectiveness of control measures. Here, we identify a complementary axis of variation that has been overlooked: variation in the *timing* of individual transmission. We introduce a smoothness parameter, ψ , that captures how much of the generation interval’s variability arises within versus between individuals—that is, whether a person’s infectiousness is concentrated in a brief burst or spread over a broad window. This quantity is invisible to standard epidemic models: the basic reproduction number R_0 , the generation interval distribution $g(\tau)$, and the population-level infectiousness profile $A(\tau)$ are all invariant to ψ . Yet its effects on stochastic dynamics can be substantial.

Our central finding is that narrow individual infectiousness profiles generate overdispersion in secondary infection counts through a mechanism that is fundamentally distinct from previously recognized sources of superspreading. When a person’s infectiousness is concentrated in a short window, the overlap between that window and their time-varying contact pattern becomes a matter of chance—producing “bad luck” superspreading that cannot be predicted or targeted by identifying high-risk individuals. We derive a closed-form expression for the resulting overdispersion and show that it depends on the interaction between infectiousness timing and the timescale of contact variation.

Using transmission-tree data from eight pathogens, we estimate ψ empirically, finding values ranging from near zero (measles, pneumonic plague) to near one (COVID-19, smallpox)—evidence that this axis of variation is not merely theoretical but differs meaningfully across pathogens. We further show that ψ governs the sensitivity of detect-and-isolate interventions to shifts in detection timing, modulates the effects of gathering size restrictions beyond their impact on R_{eff} , inflates early-epidemic stochasticity, and reduces the effective information content of serial interval data for generation interval estimation.

We believe this work is appropriate for *Nature* for several reasons. First, it identifies a genuinely new mechanism of superspreading—a topic of broad interest since the SARS and COVID-19 pandemics—that operates even in populations with no biological or behavioural heterogeneity.

Second, it introduces a parameter (ψ) that fills a gap in the field’s conceptual toolkit, complementing R_0 and k as a fundamental descriptor of transmission dynamics. Third, the findings have immediate practical implications: they reveal that evaluating interventions purely in terms of R_{eff} can be misleading, and they identify specific ways in which contact-tracing data collection and epidemic modelling practice should change. The paper combines analytic theory, simulation, and empirical estimation across multiple pathogens, in a structure parallel to Lloyd-Smith et al. (2005).

This manuscript has not been published or submitted elsewhere. All authors have approved the manuscript for submission. We have no competing interests to declare.

We suggest the following reviewers, who have expertise in individual-level transmission dynamics, generation intervals, and superspreading:

- Jamie Lloyd-Smith (UCLA) — individual variation in infectiousness, superspreading
- Jonathan Dushoff (McMaster University) — generation intervals, individual-level transmission kernels
- Sebastian Funk (London School of Hygiene and Tropical Medicine) — real-time epidemic modelling, renewal equation methods
- Christophe Fraser (University of Oxford) — contact tracing, individual-level transmission dynamics

Thank you for considering our manuscript.

Sincerely,

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